MAE339 – Aerospace Engineering Lab 4

Lift and Drag on Bodies in a Wind Tunnel

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Introduction

The purpose of this lab was to perform lift and drag performance of different bodies in a wind tunnel. Shapes like spheres (of different surface finishes) and a scaled down NACA 4412 (10.16 cm chord length) were used under varying wind conditions to observe the pressure and drag effects on each body.

Methods & Theories

Lab 4 was split into 3 main parts.

Part 1 – Measuring the free-stream velocity

- This experiment had us mainly focus on gathering free-stream pressure data in the wind tunnel with the fan on, but no bodies installed.
- The fan was started at 10Hz frequency, which was increased at an interval of 5Hz until 35Hz were reached.
- The free-stream pressure data was gathered and the free-stream velocities were calculated for each fan frequency using the following equation:

$$q_\infty = \frac{1}{2}\rho_\infty V_\infty^2$$

Part 2 – Aerodynamic Drag on a Sphere

- A force balance equipment connected with a DAQ was installed in the experiment section of the wind tunnel.
- The Spheres (rough and smooth) were sequentially installed on the force balance and the fan frequency was increased in the same manner as it was in part 1.
- Drag and lift values were noted down from the DAQ after they were 'zeroed' in properly before starting the experiment with each sphere
- The Reynold number (Re) and the coefficient of drag (C_d) on the sphere was calculated for each speed value from the equations provided below.

$$R_e = \frac{U_\infty D}{v_\infty} \qquad F_D = \frac{1}{2}c_D \rho A V^2$$

- The respective C_d and Re values were plotted accordingly via MATLAB. Plot will be provided in the next section

Part 3 – NACA 4412 by Surface Pressure Distribution

- This part focused on the surface pressure distribution experienced by a NACA 4412 airfoil under different angles of attack under a constant 35Hz input speed for the wind tunnel fan.
- The Angles were adjusted in intervals at -7°, -3°, 0°, +3°, +7°, +13°.
- Pressure values were collected at each angle of attack and the airfoil was locked in place until all the needed values were noted down adequately.
- Numerical integration function (Trapezoidal rule) was employed as it is a built-in function ('trapz()') in MATLAB for the ease of data processing and reducing human error.
- Tap locations were normalized and C_l, C_f,,C_pl, and C_pu values were calculated using the provided equations.
- Once the C_l values were calculated, they were plotted against angles of attack along with the C_l values of the same angles from airfoiltools.com.
- After processing this, the coefficient of moments C_m (leading edge), ζ(CP), C_m (c/4), m_0, ζ(AC), and C_m (AC) was calculated at each angle of attack and were then plotted.
- The following equations were used to calculate the required values:

$$\frac{P_l - P_u}{q_\infty} = \frac{P_l - P_\infty}{q_\infty} - \frac{P_u - P_\infty}{q_\infty} = C_{P,l} - C_{P,u} \qquad m_0 = \frac{dC_{M,c/4}}{d\alpha}$$

$$C_F = \int_0^c \frac{P_l - P_u}{q_\infty}\, d\left(\frac{x}{c}\right) = \int_0^1 C_{P,l} - C_{P,u}\, d\xi \qquad \xi_{CP} = -\frac{C_{M,LE}}{C_F}$$

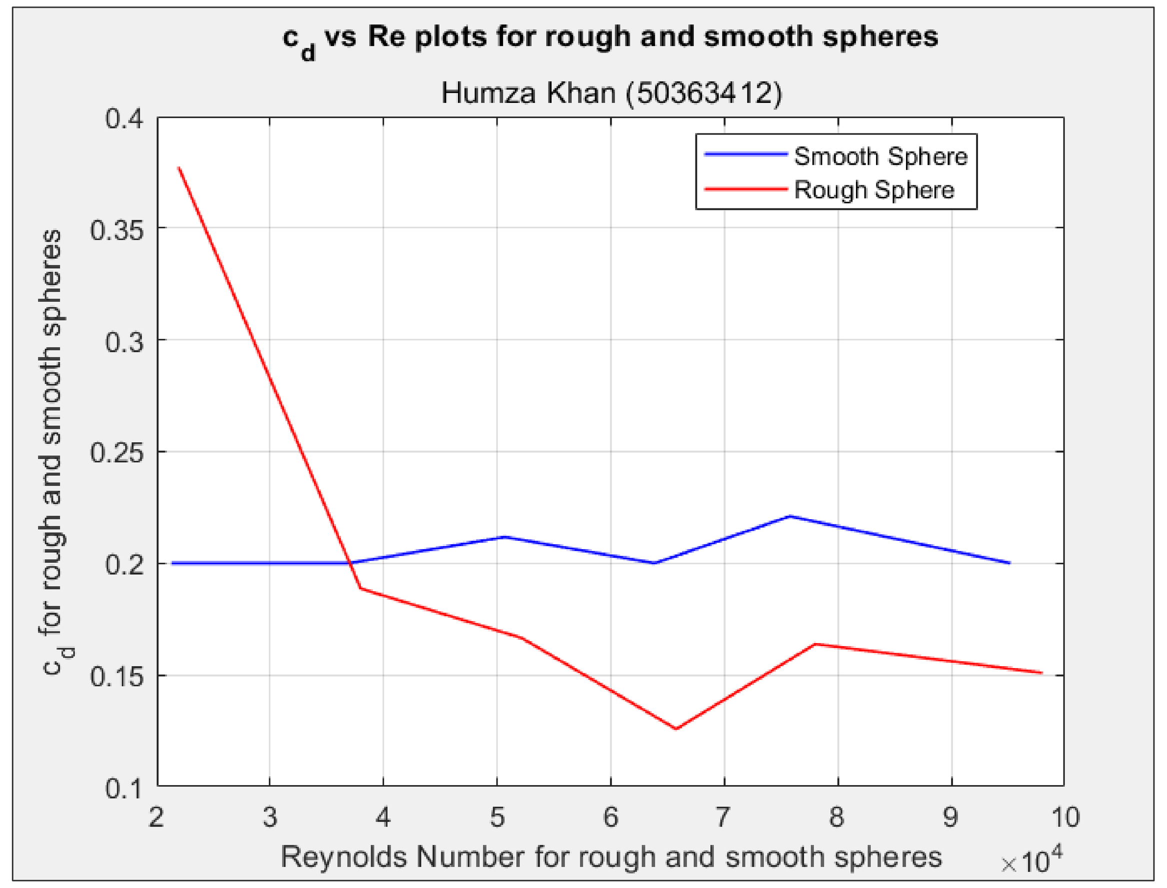
$$C_{M,LE} = \int_0^c \left(\frac{x}{c}\right)\frac{P_l - P_u}{q_\infty}\, d\left(\frac{x}{c}\right) = -\int_0^1 \xi\left(C_{P,l} - C_{P,u}\right) d\xi$$

$$C_M = -(\xi_{CP} - \xi)C_F \qquad \xi_{AC} = \xi_{CP} + \frac{C_{M,AC}}{C_F}$$

$$\xi_{AC} = -\frac{m_0}{2\pi} + 0.25$$

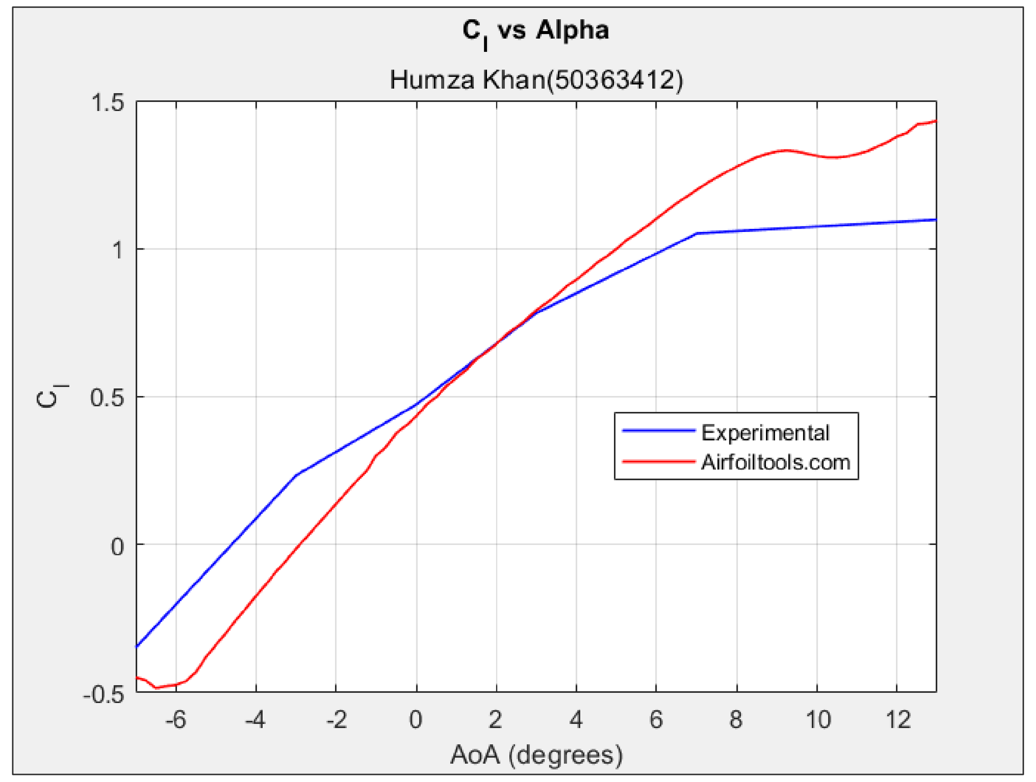
Results

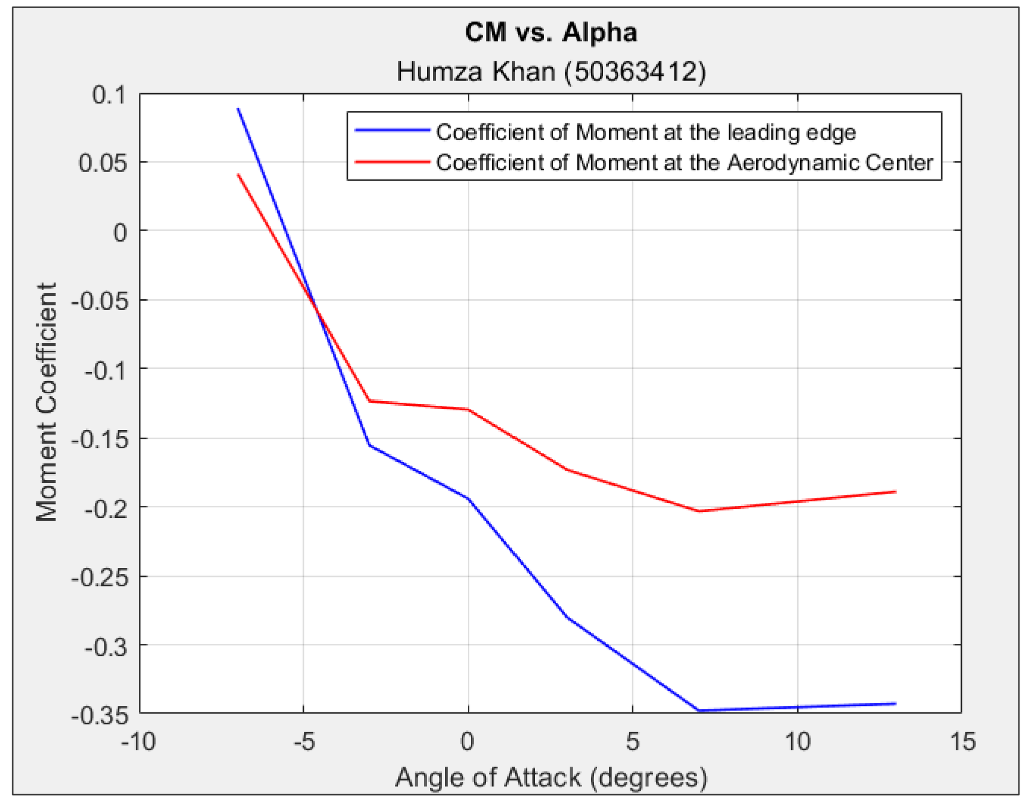
After processing the data, all the data was delivered plotted as required as the following.



From all the generated plots, the following can be derived.

- **That the smooth sphere has a more consistent C_d value when compared to the rough sphere, which fluctuates a lot.**
- **C_l vs Alpha plot were quite close enough when values airfoiltools.com and the experiment were compared. It can be safely deduced that the airfoil will stall at about 8° or 9° angle of attack**
- **C_m(LE) before stall implies aircraft stability. A negative slope implies the tendency of the airfoil to return to its stable position rather than flipping away uncontrollably and becoming completely unpredictable.**





Conclusion

The following things can be concluded from these experiments;

- Rough surfaces produce unpredictable, random drag that can lead to instability and inefficiency.
- The conditions and results of the experiment can be safely scaled to larger prototypes as the outcome was very close the real, authentic data from arifoilstool.com.
- A negative slope of the C_m vs alpha indicates stability, as the airfoil wants to naturally return to its equilibrium position.

I also want to add that there was a lot of calculations done on MATLAB but all of that cannot be neatly presented on the poster as it will require a lot more space than is available.